Laboratory Work 6 Activity

Detecting Anomalies in Time-Series Data Using Power BI

Part 1. Data Preparation

1. Download the dataset and open Power BI Desktop.

2. Click Get Data → Text/CSV and import the dataset.

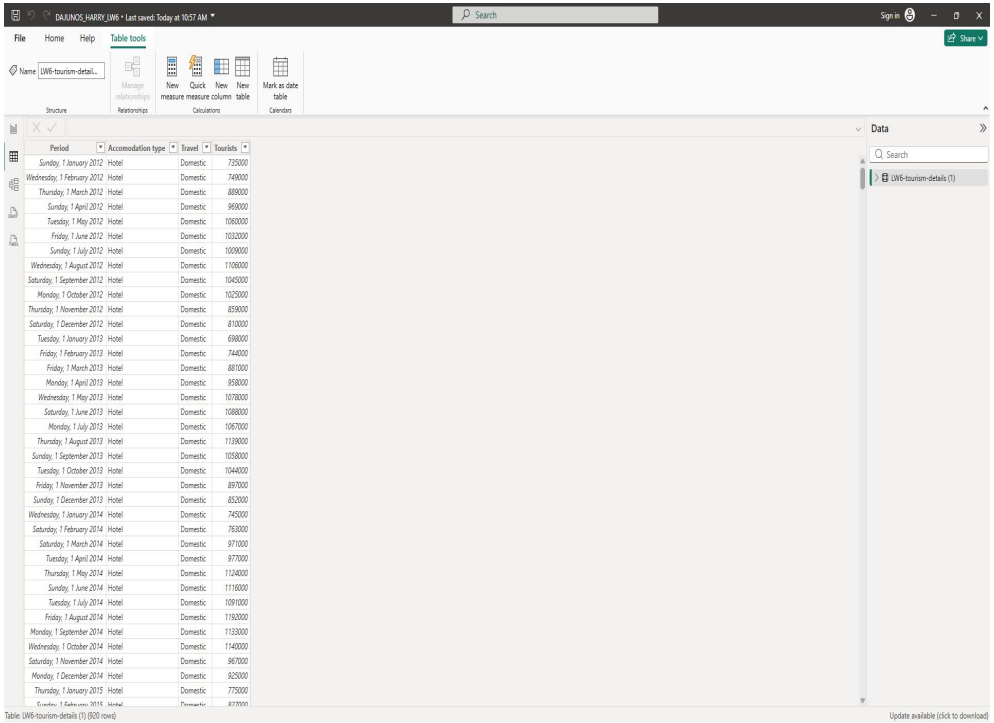
3. Ensure the following fields are correctly assigned:

○ Period → Date

○ Accommodation Type → Text

○ Travel → Text

○ Tourists → Whole Number



Period	Accomodation type	Travel	Tourists
Sunday, 1 January 2012	Hotel	Domestic	735000
Wednesday, 1 February 2012	Hotel	Domestic	749000
Thursday, 1 March 2012	Hotel	Domestic	889000
Sunday, 1 April 2012	Hotel	Domestic	969000
Tuesday, 1 May 2012	Hotel	Domestic	1060000
Friday, 1 June 2012	Hotel	Domestic	1032000
Sunday, 1 July 2012	Hotel	Domestic	1009000
Wednesday, 1 August 2012	Hotel	Domestic	1106000
Saturday, 1 September 2012	Hotel	Domestic	1045000
Monday, 1 October 2012	Hotel	Domestic	1025000
Thursday, 1 November 2012	Hotel	Domestic	859000
Saturday, 1 December 2012	Hotel	Domestic	810000
Tuesday, 1 January 2013	Hotel	Domestic	698000
Friday, 1 February 2013	Hotel	Domestic	744000
Friday, 1 March 2013	Hotel	Domestic	881000
Monday, 1 April 2013	Hotel	Domestic	958000
Wednesday, 1 May 2013	Hotel	Domestic	1078000
Saturday, 1 June 2013	Hotel	Domestic	1088000
Monday, 1 July 2013	Hotel	Domestic	1067000
Thursday, 1 August 2013	Hotel	Domestic	1139000
Sunday, 1 September 2013	Hotel	Domestic	1058000
Tuesday, 1 October 2013	Hotel	Domestic	1044000
Friday, 1 November 2013	Hotel	Domestic	897000
Sunday, 1 December 2013	Hotel	Domestic	852000
Wednesday, 1 January 2014	Hotel	Domestic	745000
Saturday, 1 February 2014	Hotel	Domestic	763000
Saturday, 1 March 2014	Hotel	Domestic	971000
Tuesday, 1 April 2014	Hotel	Domestic	977000
Thursday, 1 May 2014	Hotel	Domestic	1124000
Sunday, 1 June 2014	Hotel	Domestic	1116000
Tuesday, 1 July 2014	Hotel	Domestic	1091000
Friday, 1 August 2014	Hotel	Domestic	1192000
Monday, 1 September 2014	Hotel	Domestic	1133000
Wednesday, 1 October 2014	Hotel	Domestic	1140000
Saturday, 1 November 2014	Hotel	Domestic	967000
Monday, 1 December 2014	Hotel	Domestic	925000
Thursday, 1 January 2015	Hotel	Domestic	775000

Part 2. Data Visualization

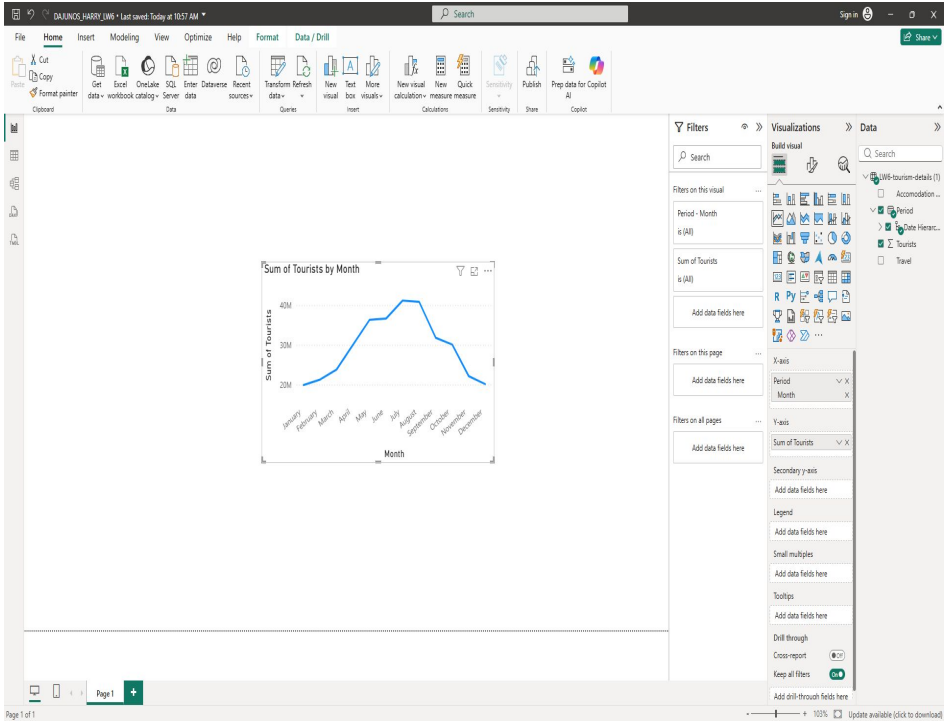
4. Insert a Line Chart visualization.

5. Drag:

○ Period → Axis

○ Tourists → Values

6. Expand the date hierarchy to show monthly data.



Part 3. Enable Anomaly Detection

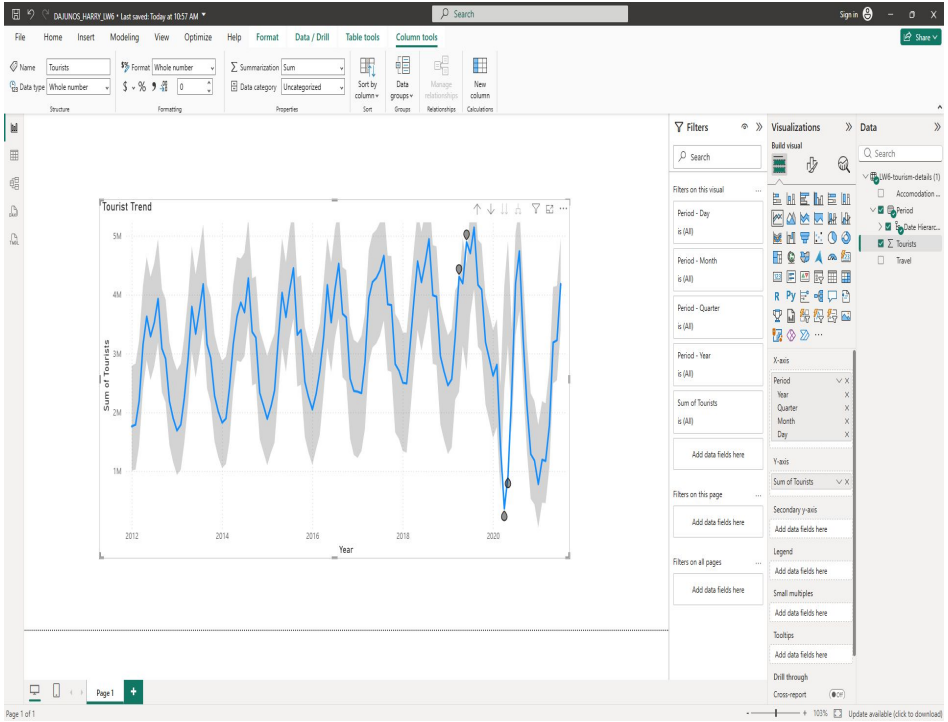
7. Go to the Analytics Pane.

8. Click Find Anomalies → Add.

9. Observe:

○ Highlighted anomaly points

○ Expected value range (shaded region)



Part 4. Explain Anomalies

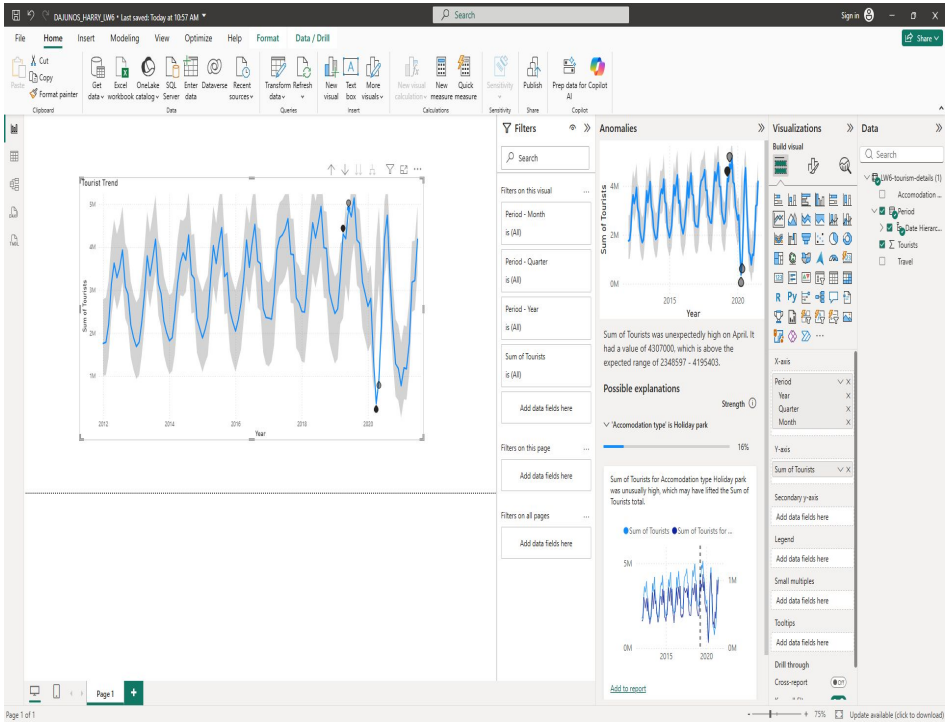
10. Drag the following into Explain By:

- Accommodation Type
- Travel

11. Click on any detected anomaly.

12. Analyze the Anomalies Pane:

- Identify whether the anomaly is high or low
- Review suggested explanations



Part 5. Interpretation and Analysis

1. Trend & Seasonality Patterns

Trend: There is a noticeable upward trend in the total number of tourists from approximately 2012 to 2019, indicating an increase in tourism over time.

Seasonality: Strong, recurring annual cycles are evident. Peaks occur consistently (likely during holiday seasons or summer months), followed by troughs in off-peak periods. This pattern remains consistent year after year, suggesting a predictable seasonal demand.

2. Obvious vs. Subtle Anomalies

Obvious anomalies: A significant decline around 2020, where tourist numbers fall drastically below the anticipated range. A sudden increase shortly before or after, where values surpass the upper confidence band.

Subtle anomalies: Minor deviations where peaks are slightly above or below expectations but still within the forecast range. Small irregularities in seasonal peaks that do not perfectly align with previous years.

3. Factors Contributing to Anomalies

The anomaly panel indicates: A spike (e.g., in April) is associated with higher values in a specific category, such as “Holiday park” accommodation type.

Likely real-world contributors include: Major disruptions (e.g., global events like travel restrictions in 2020). Changes in travel preferences, such as a rise in demand for certain types of accommodation. External factors like economic conditions, weather, or policy changes.

4. Are the Explanations Logically Valid?

Partially valid: The assertion that a specific category (like holiday parks) contributed to higher totals is mathematically sound—if one segment experiences a spike, the total will increase. However, it is incomplete: It does not clarify why that category saw an increase. For instance: Was it a result of travel restrictions encouraging local stays?

Seasonal promotions or events?

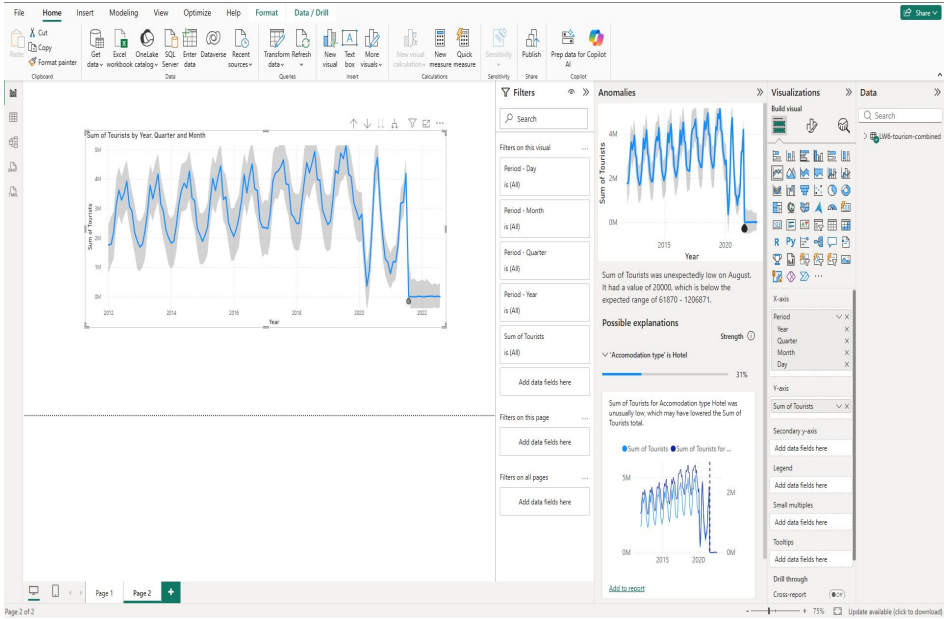
Part 6. Enhancement Task

14. Add your own new anomaly dataset (sample below) into your existing dataset.

15. Refresh the report and observe changes.

16. Compare:

- Before vs. after anomaly detection
- Newly detected anomalies



Expected Output

Anomaly Detection Analysis: Before vs. After

The first dashboard (Page 1) shows the Tourist Trend using the original dataset (LW6-tourism-details), while the second (Page 2) presents the Sum of Tourists by Year, Quarter, and Month from a combined dataset (LW6-tourism-combined), with anomaly detection applied in both cases.

In the first image, the anomaly panel flagged an unexpectedly high value in April, recording 4,307,000 tourists surpassing the expected range of 2,348,597 to 4,195,403. The likely explanation pointed to Holiday park accommodations as a contributing factor, with a relatively low explanatory strength of 16%.

In the second image, the anomaly detected was the opposite, an unexpectedly low value in August, with only 20,000 tourists recorded, far below the expected range of 61,870 to 1,206,871. This anomaly coincided with the sharp visible drop around 2020–2021, most likely reflecting the impact of the COVID-19 pandemic on tourism. The primary explanation here was Hotel accommodation, carrying a much stronger explanatory strength of 31%.

Comparing both visuals, the overall tourist trend follows a consistent seasonal oscillation from 2012 to around 2019, after which a dramatic decline disrupts the pattern. The after-detection view (Page 2) captures a more severe and prolonged dip, suggesting the combined dataset extends further in time and reveals the pandemic's full impact more clearly. Together, both anomalies highlight how accommodation type plays a significant role in explaining unusual tourist volume shifts.